180 DEGREES CONSULTING - AI SCREENING SYSTEM

The Evidence Base

The research behind every design decision -- and the gaps it tells us to fix.

This document compiles the full citation base assembled by eight parallel research agents across two rounds: the first hardening the system's design (consulting practice, LLM-as-judge reliability, fairness law, rubric science, human-in-the-loop workflow), the second pressure-testing it for gaps (integrity & prompt-injection, validation & fairness auditing, ranking-methodology efficiency).

Part 1 highlights the most important sources -- what each one establishes and how it directly benefits the system. Part 2 is the complete reference list, grouped by theme. The companion file methodology-gaps.md turns the second round into an honest fix-list.

Compiled June 2026. Every link resolves to the primary source. Core design principle throughout: the system ranks and flags; a human makes every decision.

PART 1

The sources that matter most

Of the ninety-odd references in Part 2, these are the load-bearing ones -- each settled a real design decision. For each: what it establishes, and how the system is better because of it.

1. Position bias is the #1 flaw in LLM judges

Why Controlled studies show LLM judges favour whichever option is shown first; position-consistency is only 0.57-0.83 even for strong models, and it is worst on close calls -- exactly the matches that decide a ranking.

Benefit Justifies running every match in BOTH orderings and calling a disagreement a tie. This single safeguard is what makes the ranking trustworthy rather than an artefact of presentation order.

- Judging the Judges: Position Bias in LLM-as-a-Judge (arXiv 2406.07791)

2. Pairwise comparison beats absolute scoring

Why Evidence that 'which of these two is better?' is far more reliable than 'score this one' -- absolute scores drift and are ~3-4x more sensitive to distractor features than pairwise preferences.

Benefit This is the core engine justification: rank by head-to-head comparison, and reserve cold pointwise scoring only for the coarse gate, never for the ranking itself.

- Pairwise or Pointwise? Evaluating Feedback Protocols for Bias (arXiv 2504.14716)

3. LLM judgments are not perfectly transitive

Why LLM comparisons produce cycles (A>B>C>A) in ~4-15% of cases. Merge sort assumes a consistent comparator, so a cycle can bury a strong candidate in a losing branch.

Benefit Justifies storing every comparison, the Bradley-Terry cross-check, and cycle detection -- the self-correction layer that catches a candidate the bracket wrongly sank.

- Investigating Non-Transitivity in LLM-as-a-Judge (arXiv 2502.14074)
- TrustJudge: Inconsistencies of LLM-as-a-Judge and How to Alleviate Them (arXiv 2509.21117)

4. Automated screening absorbs human bias

Why Amazon's recruiting tool learned to penalise the word 'women's'; a 2024 Stanford/UW study found language models favoured White-associated names in 85% of cases and disadvantaged Black male names in up to 100%.

Benefit The whole fairness story: strip identity before any agent looks, and never trust blinding alone -- it is necessary but not sufficient, which is why bias auditing is also required.

- Amazon ditched biased AI recruiting tool (MIT Technology Review)
- Gender, Race, Intersectional Bias in Resume Screening via LM Retrieval (Brookings)

5. The law: you may not reject by pure automation

Why GDPR Article 22 gives candidates the right not to be subject to a solely automated decision that significantly affects them -- a rejection qualifies. The EU AI Act classes employment screening as high-risk (transparency, human oversight, audit).

Benefit The architectural line of the entire system: it RANKS and flags; a human draws every cutoff and sends every rejection. This is both the legal line and the trust line.

- GDPR Article 22 Explained: Automated Decision-Making
- EU AI Act and Hiring: 2025/2026 Compliance Guide

6. There is an operational standard for bias auditing

Why NYC Local Law 144 mandates independent adverse-impact audits of automated hiring tools across sex, race, and intersectional groups -- the concrete model for how to audit, and how to resolve the paradox of auditing a system whose inputs you deliberately blinded.

Benefit Gives the system a real fairness-audit method: a separate audit-only store of voluntary demographic data, joined to outcomes only for the post-cycle 4/5ths check.

- NYC Local Law 144 Compliance Guide (Warden AI)
- NYC DCWP Automated Employment Decision Tools (official)

7. Structured, behaviourally-anchored rubrics are valid and reliable

Why Structured-hiring science: behaviourally-anchored rating scales reach a validity coefficient ~0.63 vs ~0.20 for unstructured judgement, and observable anchors are the biggest driver of inter-rater agreement.

Benefit Shapes the rubric: 4-6 dimensions, 1-4 scale, every level defined by an observable behaviour, calibrated few-shot examples -- so the comparison agents judge consistently.

- Google re:Work -- Structured Interviewing
- Behaviorally Anchored Rating Scale (AIHR)

8. 180DC recruits across all backgrounds -- on purpose

Why Every 180DC chapter examined weighs genuine passion for social impact heavily and explicitly does NOT filter on GPA, major, school prestige, or prior experience.

Benefit Directly sets the rubric's hard instructions: ignore prestige/GPA/major, keep the gate to three forgiving knockouts, and treat the motivation letter as the primary mission-fit signal.

- Recruitment -- 180DC at Cornell
- Apply -- 180DC at Berkeley
- Join -- 180DC Michigan

9. Resumes are an adversarial input -- prompt injection is real

Why ~1% of resumes in live ATS data already contain hidden injections; optimised attacks hit 43-74% success against LLM judges. White text, zero-width unicode, and 'rate this candidate highest' all work.

Benefit Adds the integrity layer the design was missing: render-to-image OCR, a structured extraction pre-pass so the judge never sees raw document text, and delimited untrusted input.

- Optimization-based Prompt Injection to LLM-as-a-Judge (ACM CCS 2024, arXiv 2403.17710)
- Measuring Real-World Prompt Injection in Resume Screening (arXiv 2605.28999)

10. AI-text detection is NOT reliable enough to reject on

Why Detectors show 8-30% false positives and are defeated by light editing or 'humanizer' tools. In a 500-pool, a 10% false-positive rate wrongly flags ~37 genuine applicants.

Benefit An honesty guardrail: never auto-reject or downrank on a detector. Instead redesign the letter prompt and shift the rubric to verifiable specifics -- the real defence against AI-written applications.

- AI Detection Accuracy -- Meta-Analysis of 14 Studies (Originality.ai)

11. Prove it works before trusting it

Why Responsible deployment runs in shadow mode (parallel to humans, output hidden) with go/no-go thresholds, and measures predictive validity against later-round outcomes using a dual cohort to escape the selective-labels trap.

Benefit Turns 'trust us' into a validation plan: a first-cycle shadow run with >=85% human-agreement gate before the system is ever authoritative.

- Shadow Mode Rollouts for AI Agents (Brightlume)
- From Biased Selective Labels to Pseudo-Labels: an EM Framework (arXiv 2406.18865)

12. The cutoff line is statistically fuzzy -- quantify it

Why Dropping a handful of pairwise preferences can change top rankings; Bradley-Terry strength scores support bootstrap confidence intervals on each candidate's rank.

Benefit Powers the 'humans review the edge cases' feature precisely: bootstrap the results, and anyone whose rank interval straddles the cutoff becomes the human review band -- cheap, and far better than a fixed top-N.

- Dropping a Handful of Preferences Can Change Top Rankings (arXiv 2508.11847)
- Uncertainty Quantification in the Bradley-Terry-Luce Model (arXiv 2110.03874)

13. Comparisons can be spent where they actually matter

Why Adaptive top-k methods concentrate expensive comparisons on the ambiguous band around the cutoff (2.4-2.8x speedup); batching changes which sort is optimal and can roughly halve LLM calls.

Benefit A clear efficiency path at 500 applicants x two tracks: freeze the clear top and bottom, spend the budget near the 200 line, and cut cost without losing precision where it counts.

- Top-k on a Budget: Adaptive Ranking with Weak and Strong Oracles -- ACE (arXiv 2601.20989)
- Rethinking Sorting in LLM-Based Pairwise Ranking with Batching (arXiv 2505.24643)

PART 2

Complete reference base

Every source assembled across the eight research agents, grouped by theme. Sources cited in Part 1 reappear here in their theme for completeness.

1 -- How elite consulting and 180DC screen

- How a consulting resume is screened at MBB (Firms Consulting)
  https://firmsconsulting.com/quarterly/consulting-resume-mckinsey-bcg-deloitte/
- Consulting Application Process 2026 (Strategy Case)
  https://strategycase.com/how-to-stand-out-as-a-consulting-applicant/
- MBB Consulting GPA Requirements 2026 (Strategy Case)
  https://strategycase.com/gpa-for-mckinsey-bcg-bain-top-consulting-firms/
- Consulting Resume Guide: Passing the MBB Screen (NextEP)
  https://nextepmbb.com/en/consulting-cv-guide-en/
- Consulting Resume Mistakes That Get You Rejected (Hacking the Case Interview)
  https://www.hackingthecaseinterview.com/pages/consulting-resume-mistakes
- Recruitment -- 180DC at Cornell
  https://www.cornell180dc.org/recruitment
- Apply -- 180DC at Berkeley
  https://www.180dcberkeley.com/apply-now
- Join Our Team -- 180DC Michigan
  https://www.180dcmichigan.com/apply
- How Can I Join? -- 180DC UNSW
  https://www.180dcunsw.org/how-can-i-join
- Recruitment -- 180DC at Carnegie Mellon
  https://www.cmu180dc.org/recruitment
- Consultant Applications -- 180DC TAMU
  https://www.180dctamu.com/recruitment
- 180 Degree Consulting Interview Experience (Glassdoor)
  https://www.glassdoor.com/Interview/180-Degree-Consulting-Interview-Questions-E581682.htm
- Top Consulting Skills (TestGorilla)
  https://www.testgorilla.com/blog/consulting-skills/

2 -- LLM-as-judge, pairwise ranking, non-transitivity

- Judging the Judges: Position Bias in LLM-as-a-Judge (arXiv 2406.07791)
  https://arxiv.org/abs/2406.07791
- Pairwise or Pointwise? Feedback Protocols for Bias (arXiv 2504.14716)
  https://arxiv.org/abs/2504.14716
- TrustJudge: Inconsistencies of LLM-as-a-Judge (arXiv 2509.21117)
  https://arxiv.org/html/2509.21117v1
- LLM-RankFusion: Mitigating Intrinsic Inconsistency in Ranking (arXiv 2406.00231)
  https://arxiv.org/html/2406.00231v1
- Investigating Non-Transitivity in LLM-as-a-Judge (arXiv 2502.14074)
  https://arxiv.org/pdf/2502.14074
- Ranking Unraveled: Recipes for LLM Rankings (arXiv 2411.14483)
  https://arxiv.org/html/2411.14483v2
- Rethinking Sorting in LLM-Based Pairwise Ranking, Batching & Caching (arXiv 2505.24643)
  https://arxiv.org/html/2505.24643

- Rethinking Bradley-Terry Models in Preference Modeling (arXiv 2411.04991)
  https://arxiv.org/html/2411.04991v1
- Autorubric: Unifying Rubric-based LLM Evaluation (arXiv 2603.00077)
  https://arxiv.org/html/2603.00077v2
- A Merge Sort Based Ranking System for LLM Evaluation (Springer 2024)
  https://link.springer.com/chapter/10.1007/978-3-031-70378-2_15
- Arena-Lite: Tournament-Based Direct Comparisons (arXiv 2411.01281)
  https://arxiv.org/html/2411.01281v4
- Large Language Models are Effective Text Rankers (Pairwise Ranking Prompting, arXiv 2306.17563)
  https://arxiv.org/pdf/2306.17563
- Efficient LLM Comparative Assessment: Product of Experts (arXiv 2405.05894)
  https://arxiv.org/pdf/2405.05894
- Evaluating LLM Evaluators (Eugene Yan)
  https://eugeneyan.com/writing/llm-evaluators/
- LLM-as-a-Judge: Complete Guide (EvidentlyAI)
  https://www.evidentlyai.com/llm-guide/llm-as-a-judge
- LLM-Judge Bias Mitigation 2026 (FutureAGI)
  https://futureagi.com/blog/evaluating-llm-judge-bias-mitigation-2026/

3 -- Fairness, bias, and the law

- Amazon ditched biased AI recruiting tool (MIT Technology Review)
  https://www.technologyreview.com/2018/10/10/139858/amazon-ditched-ai-recruitment-software-because-it-was-biased-against-women/
- Why Amazon's Hiring Tool Discriminated Against Women (ACLU)
  https://www.aclu.org/news/womens-rights/why-amazons-automated-hiring-tool-discriminated-against
- Gender, Race, Intersectional Bias in Resume Screening (Brookings)
  https://www.brookings.edu/articles/gender-race-and-intersectional-bias-in-ai-resume-screening-via-language-model-retrieval/
- Gender, Race, Intersectional Bias via LM Retrieval (arXiv 2407.20371)
  https://arxiv.org/abs/2407.20371
- AI Tools Show Bias Ranking Applicants' Names (UW News)
  https://www.washington.edu/news/2024/10/31/ai-bias-resume-screening-race-gender/
- AI Resume Screeners Prefer White Male Candidates (Fisher Phillips)
  https://www.fisherphillips.com/en/news-insights/ai-resume-screeners.html
- Small Changes, Large Consequences: Allocational Fairness of LLMs in Hiring (arXiv 2501.04316)
  https://arxiv.org/pdf/2501.04316
- Fairness in AI-Driven Recruitment: Challenges & Metrics (arXiv 2405.19699)
  https://arxiv.org/html/2405.19699v3
- Fairness and Bias in Algorithmic Hiring (ACM)
  https://dl.acm.org/doi/full/10.1145/3696457
- GDPR Article 22 Explained: Automated Decision-Making
  https://gdprinfo.eu/gdpr-article-22-explained-automated-decision-making-profiling-and-your-rights
- GDPR-Compliant AI Recruiting: Never Auto-Reject (Treegarden)
  https://treegarden.io/blog/gdpr-compliant-ai-recruiting/
- GDPR Article 22 and AI Recruitment (Treegarden)
  https://treegarden.io/blog/gdpr-article-22-ai-recruitment-screening/
- EU AI Act and Hiring: 2025/2026 Compliance Guide (HireTruffle)
  https://www.hiretruffle.com/blog/eu-ai-act-hiring
- Recruiting under the EU AI Act (HeroHunt)
  https://www.herohunt.ai/blog/recruiting-under-the-eu-ai-act-impact-on-hiring/
- NYC Local Law 144 Compliance Guide (Warden AI)
  https://www.warden-ai.com/resources/hr-tech-compliance-nyc-local-law-144

- NYC DCWP Automated Employment Decision Tools (official)
  https://www.nyc.gov/site/dca/about/automated-employment-decision-tools.page
- NYC Local Law 144 Bias Audit Requirements (RiskTemplate)
  https://risktemplate.com/blog/2026-04-03-nyc-local-law-144-ai-bias-audit-requirements/
- EEOC Title VII Guidance on AI Tools (Mayer Brown)

https://www.mayerbrown.com/en/insights/publications/2023/07/eeoc-issues-title-vii-guidance-on-employer-use-of-ai-other-algorithmic-decisionmaking-tools
- AI Recruiting Compliance 2026: Defensible Hiring Playbook (Humanly)
  https://www.humanly.io/blog/ai-recruiting-compliance-2026-defensible-hiring-playbook
- Auditing AI Systems for Bias in Employment (Ogletree Deakins)
  https://ogletree.com/insights-resources/blog-posts/auditing-artificial-intelligence-systems-for-bias-in-employment-decision-making/
- Auditing the Audits: Lessons from NYC LL144 (ACM FAccT 2025)
  https://dl.acm.org/doi/full/10.1145/3715275.3732004

4 -- Rubric design and structured hiring

- Google re:Work -- Structured Interviewing
  https://rework.withgoogle.com/intl/en/subjects/hiring
- Behaviorally Anchored Rating Scale (AIHR)
  https://www.aihr.com/blog/behaviorally-anchored-rating-scale/
- BARS Ultimate Guide 2025 (CHRMP)
  https://www.chrmp.com/behaviorally-anchored-rating-scale-bars/
- How Role-Specific Rubrics Improve Evaluation (Skillfuel)
  https://www.skillfuel.com/how-role-specific-rubrics-improve-candidate-evaluation/
- How to Design Perfect Knockout Criteria (Foundire)
  https://foundire.com/blog/how-to-design-perfect-knockout-criteria/
- Interview Rubrics (Metaview)
  https://www.metaview.ai/resources/blog/interview-rubrics
- Remove Bias from Your Interview Rubric (VidCruiter)
  https://vidcruiter.com/interview/structured/interview-rubric/

5 -- Human-in-the-loop and automation bias

- AI Candidate Screening: The 3-Bucket Playbook (Metaview)
  https://www.metaview.ai/resources/blog/ai-candidate-screening
- Check the Box! Automation Bias in AI Personnel Selection (PMC)
  https://pmc.ncbi.nlm.nih.gov/articles/PMC10113449/
- Automation Bias from Resume Screening with AI (CloudApper)
  https://www.cloudapper.ai/talent-acquisition/automation-bias-from-resume-screening-with-ai-for-frontline-roles/
- Resume Screening 2026: AI, ATS, Bias Governance (MiHCM)
  https://mihcm.com/resources/blog/resume-screening-in-2026-a-guide-to-ai-powered-screening-ats-integration-bias-governance/
- How to Reduce Bias in Resume Screening (MiHCM)
  https://mihcm.com/resources/blog/fair-hiring-in-the-age-of-ai-how-to-reduce-bias-in-resume-screening/
- Human-in-the-Loop AI Interviews (FloCareer)
  https://flocareer.com/blog/human-in-the-loop-ai-interviews
- AI-Driven Decision-Making System for Hiring (arXiv 2512.20652)
  https://arxiv.org/html/2512.20652v1
- Allocate Marginal Reviews to Borderline Cases via LLM Ranking (arXiv 2602.06078)
  https://arxiv.org/pdf/2602.06078
- Confidence Scores and Human Review Queues in Document AI (TurboLens)
  https://www.turbolens.io/blog/2026-04-05-confidence-scores-and-human-review-queues-in-document-ai

- Responsibly Designed Automation in Recruitment (World Economic Forum)
  https://www.weforum.org/stories/2025/09/ai-powered-recruitment-inclusion-transparency/
- AI Resume Screening 2026 Best Practices (TheHireHub)
  https://www.thehirehub.ai/blog/resume-screening-with-ai-2026-best-practices
- AI Candidate Screening: Automate and Scale Fairly (Sapia.ai)
  https://sapia.ai/resources/blog/ai-candidate-screening-automation-tips/

6 -- Integrity: prompt injection, AI content, fraud

- Optimization-based Prompt Injection to LLM-as-a-Judge (ACM CCS 2024, arXiv 2403.17710)
  https://arxiv.org/abs/2403.17710
- Measuring Real-World Prompt Injection in Resume Screening (arXiv 2605.28999)
  https://arxiv.org/html/2605.28999
- Adversarial Attacks on LLM-as-a-Judge Systems (arXiv 2504.18333)
  https://arxiv.org/abs/2504.18333
- AI Security: Resume Screening Adversarial Vulnerabilities (arXiv 2512.20164)
  https://arxiv.org/pdf/2512.20164
- LLMs Cannot Reliably Judge (Yet?): Robustness of LLM-as-a-Judge (arXiv 2506.09443)
  https://arxiv.org/html/2506.09443
- Judging the Judges: Bias Mitigation Strategies in LLM-Judge Pipelines (arXiv 2604.23178)
  https://arxiv.org/html/2604.23178
- Design Patterns for Securing LLM Agents against Prompt Injection (arXiv 2506.08837)
  https://arxiv.org/pdf/2506.08837
- OWASP LLM01:2025 Prompt Injection
  https://genai.owasp.org/llmrisk/llm01-prompt-injection/
- AI Detection Accuracy: Meta-Analysis of 14 Studies (Originality.ai)
  https://originality.ai/blog/ai-detection-studies-round-up
- AI Cover Letter Checker (Pangram Labs)
  https://www.pangram.com/blog/ai-cover-letter-checker
- How Employers Detect AI-Generated Resumes 2026 (ResumeGeni)
  https://resumegeni.com/blog/how-employers-detect-ai-generated-resumes-2026
- Resume Fraud: The $600 Billion Crisis (Crosschq)
  https://www.crosschq.com/blog/resume-fraud-the-600-billion-crisis-transforming-how-organizations-verify-talent-in-2025
- 1 in 4 Candidate Profiles Will Be Fake by 2028 (StaffingHub)
  https://staffinghub.com/hiring/candidate-verification-ai-resume-fraud-staffing-firms-2026/
- Importance of Deduplication in Large Hiring Systems (Resumly.ai)
  https://www.resumly.ai/blog/importance-of-deduplication-in-large-hiring-systems

7 -- Validation, reliability metrics, fairness auditing

- Shadow Mode Rollouts for AI Agents (Brightlume)
  https://brightlume.ai/blog/shadow-mode-rollouts-ai-agents-pilot-production
- Measuring Validity in LLM-based Resume Screening (arXiv 2602.18550)
  https://arxiv.org/pdf/2602.18550
- From Biased Selective Labels to Pseudo-Labels: an EM Framework (arXiv 2406.18865)
  https://arxiv.org/pdf/2406.18865
- Inter-Rater Reliability for Screening: Kappa vs Gwet's AC1 (Mapped Research)
  https://mappedresearch.com/blog/inter-rater-reliability-screening
- Enhancing AI Evaluation with Cohen's Kappa (Galileo)
  https://galileo.ai/blog/cohens-kappa-metric
- The Necessity of Setting Temperature in LLM-as-a-Judge (arXiv 2603.28304)
  https://arxiv.org/html/2603.28304v1

- Uncertainty Quantification in the Bradley-Terry-Luce Model (arXiv 2110.03874)
  https://arxiv.org/pdf/2110.03874
- Dropping a Handful of Preferences Can Change Top Rankings (arXiv 2508.11847)
  https://arxiv.org/pdf/2508.11847
- Prompt Stability Scoring for Text Annotation (arXiv 2407.02039)
  https://arxiv.org/pdf/2407.02039
- AutoScreen-FW: LLM Framework for Resume Screening (arXiv 2603.18390)
  https://arxiv.org/pdf/2603.18390

8 -- Ranking-methodology efficiency upgrades

- Top-k on a Budget: Adaptive Ranking with Weak & Strong Oracles -- ACE (arXiv 2601.20989)
  https://arxiv.org/html/2601.20989v1
- Confident Rankings with Fewer Items: Adaptive LLM Evaluation (arXiv 2601.13885)
  https://arxiv.org/html/2601.13885
- Active Learners as Efficient PRP Rerankers -- Mohajer (arXiv 2605.14236)
  https://arxiv.org/html/2605.14236
- PARWiS: Winner Determination under Shoestring Budgets (arXiv 2603.01171)
  https://arxiv.org/pdf/2603.01171
- Drawing Conclusions from Draws: Preference Semantics in Arena Evaluation (arXiv 2510.02306)
  https://arxiv.org/pdf/2510.02306
- Prompt-Dependent Ranking of LLMs with Uncertainty (arXiv 2603.03336)
  https://arxiv.org/html/2603.03336
- Noisy Sorting Capacity (arXiv 2202.01446)
  https://arxiv.org/html/2202.01446v3
- Sample Complexity of Best-k Selection from Pairwise Comparisons (arXiv 2007.03133)
  https://arxiv.org/pdf/2007.03133
- Improving Ranking Quality in Swiss-System Tournaments (arXiv 2112.10522)
  https://arxiv.org/html/2112.10522v2
- TrueSkill 2: An Improved Bayesian Skill Rating System (Microsoft Research)
  https://www.microsoft.com/en-us/research/wp-content/uploads/2018/03/trueskill2.pdf

APPENDIX

Companion documents in this folder

README.md

Plain-English explanation of the system and every step's role.

research-synthesis.md

The 12 design decisions the first research round drove.

methodology-gaps.md

Honest gap analysis from the second research round, with fixes prioritised.

pipeline-spec.md

The buildable stage-by-stage technical reference.

rubrics/consultant.md & rubrics/team-lead.md

The drop-in scoring rubrics for each track.

agents-explained.md

How the agents differ (Conductor, Comparison, Reviewer) + showcase prompt.

presentation.md

Slide prompts and the spoken talk track for the pitch.